BCI project procedure – Slow cortical potential

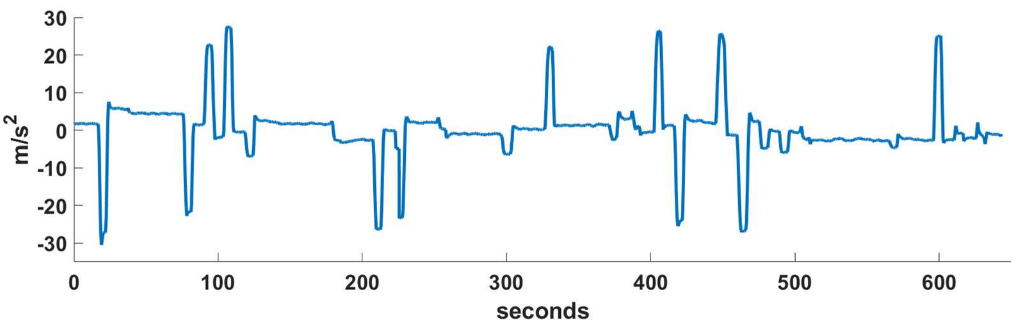
MOTOR-RELATED CORTICAL POTENTIAL

In EEG experiments, **stimulus-aligned trials** are time-locked to an external cue's onset, allowing analysis of sensory processing. This alignment captures event-related potentials (ERPs) linked to stimulus perception, such as the P300. Conversely, **response-aligned trials** synchronize data to the participant's motor response, isolating neural activity related to motor preparation and execution. Stimulus-aligned methods emphasize exogenous influences, while response-aligned approaches focus on endogenous processes.

The **motor-related cortical potential (MRCP)** is a slow negative deflection associated with movement planning. In our project, movements are **stimulus-induced**, and the MRCP begins around 500 ms after the cue. This response originates in the motor cortex and reflects preparation triggered by the external stimulus. In contrast, **self-initiated** movements (e.g., voluntary actions) generate the **Bereitschaftpotential (BP)**, starting 1-2 seconds before movement onset. The BP originates in the supplementary motor area (SMA) and reflects internally generated motor intention. Although both MRCP and BP involve activity in premotor and primary motor areas, the BP is characterized by an early onset and stronger involvement of the SMA. These features distinguish it from the stimulus-driven MRCP and highlight the neural differences between voluntary and externally cued movement planning.

Automatic detection of movement onset (optional)

- Detection of movement onset based on the information of the sensors of the exoskeleton and the data glove. Use the following channels for detecting the movement classes:
  - Channel 93: elbow flexion (1536) and extension (1537)
  - Channels 94, 95 and 96: supination (1538) and pronation (1539)
  - Channels 96 and 69: hand close (1540) and hand open (1541), respectively.
- Alternatively, you can work with the movement-onset values provided in the third column of the field "events" of the variable EEG included in the MAT files. You can also use these values to compare with your results.



An example of the channel 93 for all time samples of "subject 1 - run 1". You can see that time points with activation of the accelerometer are related with the flexion/extension movements.

Computation of the MRCP

For the calculation of the MRCP potential, all the steps of the preprocessing methodology prior to trial extraction are perfectly valid: decimation, filtering, bad channel rejection, re-referencing to CAR, ICA, and interpolation of bad channels.

- **Band-pass filtering (0.3-10 Hz) the EEG data**. The MRCP's slow dynamics (~0.3–3 Hz) necessitate low-frequency band-pass filtering to enhance signal-to-noise ratio while preserving its temporal characteristics.

  With a 512Hz sampling rate, extremely narrow bands (e.g., 0.3–3 Hz) require filter coefficients close to zero, which can lead to filtering issues. A simple solution in this case is to use a slightly wider band, such as 0.3-10 Hz. Higher frequencies (> 10 Hz) often contain noise, such as muscle artifacts, while ultra-low frequencies (<0.3 Hz) are typically affected by drift artifacts.

  (OPTIONAL) Band-pass filtering (0.3-3Hz) the EEG data. If you plan to apply a very narrow filter like the proposed one with a sample rate of 512Hz, using a simple Butterworth filter may lead to issues with the coefficients of the numerator (represented by "b", which indicate the zeros of the filter transfer function) being very close to zero. This can result in problems such as singular matrices when using the function *filtfilt*. Moreover, if the zeros are very small, it can cause the frequency response of the filter to have a prominent peak at the frequency of the zeros, which may not be desirable. Zeros near zero value can also make the filter less stable and more prone to oscillation.

  To overcome these issues, it is recommended to first decimate the EEG data, which involves resampling the data at a lower sample frequency. For example, using a factor of 16, you can downsample the data to a lower sample rate of 32Hz by applying the Matlab function *decimate*. After that, you can design the bandpass Butterworth filter using this new sample rate.

- **Trial extraction:** Construct the trials using the movement onset sample (either the one obtained in the previous (optional) step or the value from the 3rd column of the "events" field in the MAT file) along with the corresponding movement class event code. Each trial should span from -2.5 to 2.5 seconds relative to the movement onset, which is considered 0. If you're working with a 3D matrix (recommended), you should end up with a matrix of size 60 trials x 61 channels x 2560 samples for each subject and movement class. A convenient approach is to initialize this 3D matrix with NaN values to reserve space. This also allows you to mark bad trails and bad channels using NaN, keeping everything consistent and easy to handle later.

- **Trial rejection:** The idea is to apply the same algorithm or procedure designed during the preprocessing stage for trial rejection. In that stage, trials were aligned to the cue/imperative stimulus, whereas in this case they are aligned to the movement onset. However, this alignment difference is not relevant when detecting bad trials. You can use either threshold-based techniques (discarding trials where the amplitude exceeds ±150μV) or outlier detection methods based on different features such as amplitude, variance, or kurtosis.

  As a reminder, the goal of this step is to identify artifact-free trials for each channel, movement class, and subject. A simple way to organize the data is by creating a 2D matrix (60 trials x 61 channels) filled with 0s or 1s (e.g., 0 = artifact, 1 = clean). For instance, if trial $\boldsymbol{i}$ is marked as bad for channel $\boldsymbol{j}$, then the element at position $(\boldsymbol{i}, \boldsymbol{j})$ should be 0.

Remember to mark trials as bad if:

- The physical movement starts earlier than 100ms after the stimulus appears (this is considered below the minimum reaction time, suggesting the subject anticipated the cue).
- The movement starts more than 2s after the stimulus, indicating the subject likely did not respond to the stimulus.

- **Averaging**: Obtain grand-mean average motor-related potentials for each movement (mean of the good trials for each channel and mean of subjects).

  It may be interesting to compare the mean average potential for each subject with the grand-average potential across subjects. You can also compare the results between different movement classes.

- **Feature extraction:** Calculate the peak amplitude of the grand-mean average slow cortical potential.

  Note that this peak occurs before the movement onset and is typically associated with motor preparation and response readiness.

  MRCPs are commonly characterized by their peak amplitude and onset latency. To obtain a robust estimate of the peak amplitude, it is recommended to use a time window around the detected peak. Specifically, first identify the peak at channel Cz (30), then define a time window of ±100 ms centered on the detected peak time, and finally compute the average amplitude within this interval for all EEG electrodes.

- **Topographical mapping**: Plot the topographical distribution of the MRCPs using the peak amplitude values computed in the previous step for each EEG channel.

  You can use the function "topoplot.m", which is included in the EEGLAB toolbox. This function plots a 2D topographic map of scalp data (in this case, peak amplitude values), as viewed from above the head. Channel locations must be provided as an input to the function. You can find the necessary location file on Atenea (loc61eeg.ced), which includes the name of the channel of each channel along with its corresponding polar coordinates (radius and angle).

DELIVERABLES

D4. Plot the average potentials at the Cz electrode. Identify the slow cortical potential and discuss its characteristics, focusing on differences between the movement classes. Explore potential inter-subject variability.

D5. Display the topographical maps of averaged MRCP peak for each movement class. Reflect on whether the topographic distribution of cortical activity could be a useful feature for distinguishing between different types of movements.

We present two examples corresponding to grand-mean averaged MRCPs, recorded at the Cz electrode, focusing on the movement of hand closing. Additionally, we show the topographical maps of the peak amplitude of the MRCPs for this movement. These visualizations highlight the spatial distribution and temporal dynamics of brain activity during movement preparation and execution. Similar results are obtained for hand opening.

By comparing the MRCPs and their corresponding topographies for the different movement types, we aim to explore potential differences in cortical activation patterns, which may offer insights into the neural processes underlying each movement.